Problem set 1

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Table of contents

Problem Set 1

This is the first problem set. To complete this problem set, please answer questions 1 - 5, which are each located in their own file. For this homework, please try to limit yourselves to using these libraries to conduct the work:

```
library(data.table) # pick either data.table
library(dplyr)      # or dplyr; not both.
library(ggplot2)
```

1 Reason with Notation

1.1 Part (a)

In two sentences, explain the notation $Y_i(1)$.

1.2 Part (b)

In two sentences, explain the notation $Y_1(1)$.

1.3 Part (c)

In two sentences, explain the notation $E[Y_i(1)|d_i = 0]$.

1.4 Part (d)

In two sentences, explain the difference between the notation $E[Y_i(1)]$ and $E[Y_i(1)|d_i = 1]$.

2 Visual Acuity with Simple Data

```
library(data.table)
```

```
table <- data.table(
  subject = 1:7,
  y_0 = c(10, 12, 15, 11, 10, 17, 16),
  y_1 = c(12, 12, 18, 14, 15, 18, 16),
  tau = c(2, 0, 3, 3, 5, 1, 0)
)
```

2.1 Illustration

Use the values in the table below to illustrate that $E[Y_i(1)] - E[Y_i(0)] = E[Y_i(1) - Y_i(0)]$.

To do so, please write code that will index to the correct values in the table and calculate the difference between $E[Y_i(1)]$ and $E[Y_i(0)]$ for each subject. Then, calculate the average of these differences.

```
# Use this code chunk to show your code work
```

2.2 Data Possibilities

Is it possible to collect all necessary values and construct a table like the one provided in real life? Explain why or why not?

3 Visual Acuity

```r
library(data.table)
```

Suppose we are interested in the hypothesis that children playing outside leads them to have better eyesight.

Consider the following population of ten children whose visual acuity we can measure.

- Visual acuity is the decimal version of the fraction given as output in standard eye exams.
- Someone with 20/20 vision has acuity 1.0, while someone with 20/40 vision has acuity 0.5.
- Numbers greater than 1.0 are possible for people with better than “normal” visual acuity.

```r
d <- data.table(
  child = 1:10,
  y_0 = c(1.2, 0.1, 0.5, 0.8, 1.5, 2.0, 1.3, 0.7, 1.1, 1.4),
  y_1 = c(1.2, 0.7, 0.5, 0.8, 0.6, 2.0, 1.3, 0.7, 1.1, 1.4)
)
```

In this table:

- `y_1` means the measured *visual acuity* if the child were to play outside at least 10 hours per week from ages 3 to 6

- `y_0` means the measured *visual acuity* if the child were to play outside fewer than 10 hours per week from age 3 to age 6;
- Both of these potential outcomes *at the child level* would be measured at the same time, when the child is 6.

3.1 Treatment effect

Compute the individual treatment effect for each of the ten children.

```r
# Use this code chunk to show your code work
```

3.2 Story time

Tell a "story" that could explain this distribution of treatment effects. In particular, discuss what might cause some children to have different treatment effects than others.

```
# Use this code chunk to show your code work (if needed)
```

Answer: …

3.3 True ATE

For this population, what is the true average treatment effect (ATE) of playing outside.

```
# Use this code chunk to show your code work
```

Answer: …

3.4 Even-Odd split

Suppose we are able to do an experiment in which we can control the amount of time these children play outside for three years. We happen to randomly assign the odd-numbered children to treatment and the even-numbered children to control. What is the estimate of the ATE you would reach under this assignment? (Please describe your work.)

```
# Use this code chunk to show your code work
```

Answer: …

3.5 Biased or Unbiased?

How different is the estimate from the truth? In your own words, why is there a difference? Does this mean that the estimator is a biased or an unbiased estimator? Does this mean that the estimate is biased or unbiased?

```
# Use this code chunk to show your code work
```

Answer: …

3.6 How many splits are possible?

We just considered one way (odd-even) an experiment might split the children. How many different ways (every possible way) are there to split the children into a treatment versus a control group (assuming at least one person is always in the treatment group and at least one person is always in the control group)?

```
# Use this code chunk to show your code work
```

Answer: ...

3.7 Think about your assignment strategy

Given there are as many ways to assign as you answered in the last sub-question, can you provide a rationale for why you might prefer one assignment strategy over another?

For concreteness, suppose that either (a) you can have a treatment assignment where one and only one of the kids is randomly assigned to treatment; or (b) you can have a treatment assignment where exactly five of the kids are randomly assigned to treatment.

As a small hint, you might note that $\left\{ \left[\sum_{i=1}^{n} Y_i(1)|d_i = 1 \right] - \left[\sum_{j=1}^{n} Y_j(0)|d_j = 0 \right] \right\} \equiv ATE$ is an estimator and there are some properties of estimators that we care about.

To make the question tractable, suppose that if you were to increase the size of the population procedure (a) would keep a single kid in treatment, while procedure (b) would keep 50% of the sample in treatment and 50% of the sample in control.

3.8 Compute the MSE of these two designs

Because you have the entire population of kids, their entire scheduled of potential outcomes, and and two proposed sampling procedures: conduct a simulation study. First, calculate all of the possible treatment effects that you might observe under each design. Then, compute the mean-squared error of each design. Which design – the one where you have a single kid in treatment, or the one where you have five kids in treatment – produces a lower MSE? (Hint the `combn` function might help you with your subsetting.)

3.9 Observational study

Suppose that we decide it is too hard to control the behavior of the children, so we do an observational study instead. Children 1-5 choose to play an average of more than 10 hours per week from age 3 to age 6, while Children 6-10 play less than 10 hours per week. Compute the difference in means from the resulting observational data.

```
# Use this code chunk to show your code work
```

Answer: ...

3.10 Observational ATE

Compare your answer in `Observational study` to the true ATE. In your own words what causes the difference? Does this mean that the estimator is a biased or an unbiased estimator? Does this mean that the estimate is biased or unbiased?

```
# Use this code chunk to show your code work (if needed)
```

Answer: ...

4 What is an experiment, maaaan?

The following questions can be a little bit challenging. This is because the argument that you are being asked to make is based on the rote application of a definition. To begin with, it is useful for you to define what you mean when you write about either *an experiment* or *an observational study*. Then, with these definitions on hand, use the definitions to answer the following questions.

4.1 Define your terms

- **An experiment is:**
- **An experiment provides the following statistical guarantees:**
- **An observational study is:**
- **An observational study provides the following statistical guarantees:**

4.2 Does a random, iid sample produce an unbiased treatment effect estimate?

Assume that a researcher takes a random sample of elementary school children and compares the grades of those who were previously enrolled in an early childhood education program with the grades of those who were not enrolled in such a program. Is this an experiment, an observational study, or something in between?

Answer: ...

4.3 What if an official agency produces the idd sample?

Assume that the researcher works together with an organization that provides early childhood education and offer free programs to certain children. However, which children that received this offer was not randomly selected by the researcher but rather chosen by the local government. (Assume that the government did not use random assignment but instead gives the offer to students who are deemed to need it the most) The research follows up a couple of years later by comparing the elementary school grades of students offered free early childhood education to those who were not. Is this an experiment, an observational study, or something in between? Explain!

Answer: ...

4.4 What if someone else randomly assigns

If the government assigned students to treatment and control by "coin toss", rather than simply sampling the population, would you say that the study is experimental or observational? Why? What, if any guarantees does this process provide?

Answer: ...

5 Moral Panic

Suppose that a researcher finds that high school students who listen to death metal music at least once per week are more likely to perform badly on standardized test.

As a consequence, the researcher writes an opinion piece in which she recommends parents to keep their kids away from "dangerous, satanic music".

- Let the potential outcomes to control, $Y_i(0)$, be each student's test score when listening to death metal at least one time per week.
- Let $Y_i(1)$ be the test score when listening to death metal less than one time per week.

5.1 Explain the statements

Explain the statement $E[Y_i(0)|D_i = 0] = E[Y_i(0)|D_i = 1]$ in words. First, state the rote English language translation. Second, tell us the *meaning* of this statement. A full points solution will use the term "potential outcomes" twice.

Answer: ...

5.2 Can you believe it

Do you expect that this circumstance actually matches with the meaning that you've just written down? Why or why not?

Answer: ...